

EXPERIMENT - 1

Linux Programming Environment & Basic command

A) Setting up of Linux Programming Environment

We can do C programming online or offline according to the needs. It is for us to keep working on a Linux Environment.

Online: [CoCalc.ai](https://coCalc.ai) - Brief the steps followed

1. Open [CoCalc.ai](https://coCalc.ai) and sign in.
2. Create/open a project and launch a Linux terminal.
3. Create a C source file (for example, hello.c).
4. Write and save the C program.
5. Compile the program using gcc.
6. Execute the program and observe the output.

[CoCalc.ai](https://coCalc.ai) project/terminal showing the linux environment and your C program.



```
~$ vi hello.c
~$ cc hello.c -o hello
~$ ./hello
Hello World
~$ █
```

Offline Methods

Methods: Dual boot, Single boot, Bootable Pendrive, VM

Dual Boot-Two operating systems are installed on the same computer and the user selects one while starting the computer.

Single Boot-One operating system is installed and used as the main operating system.

Bootable Pendrive - A USB drive containing a bootable operating system is used to start the computer.

Virtual Machine (VM) - A virtual computer is created inside an existing operating system using virtualization software.

1. Virtual Machine Manager (hypervisor)

Type-1 (Baremetal)

A Type-1 hypervisor runs directly on the physical hardware.

Example: VMware ESXi.

Type-2 (Hosted)

A Type-2 hypervisor runs on top of a host operating system.

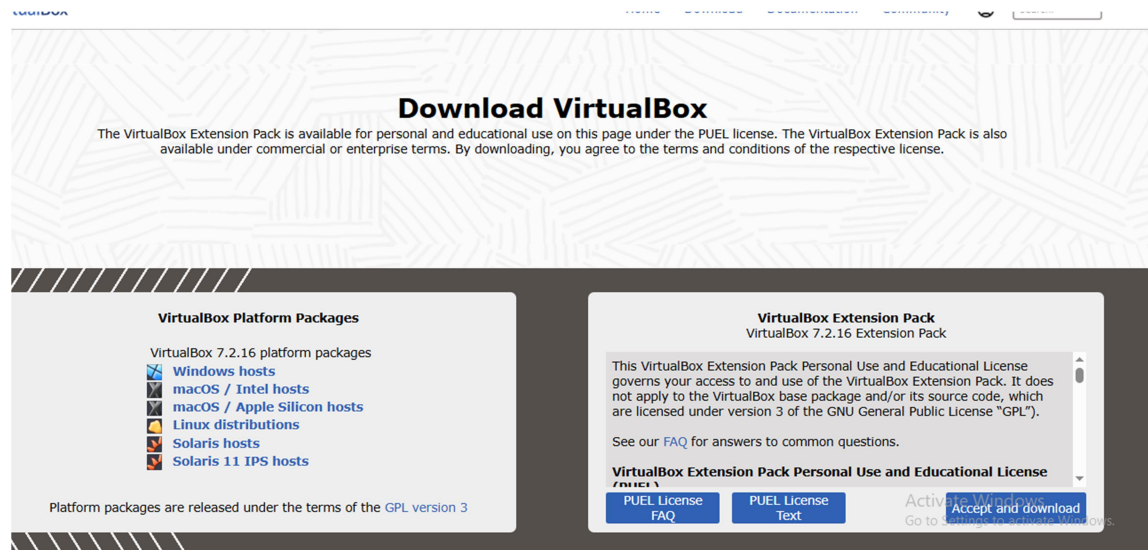
Examples: Oracle VirtualBox, VMware Workstation

We would be using Oracle VirtualBox.

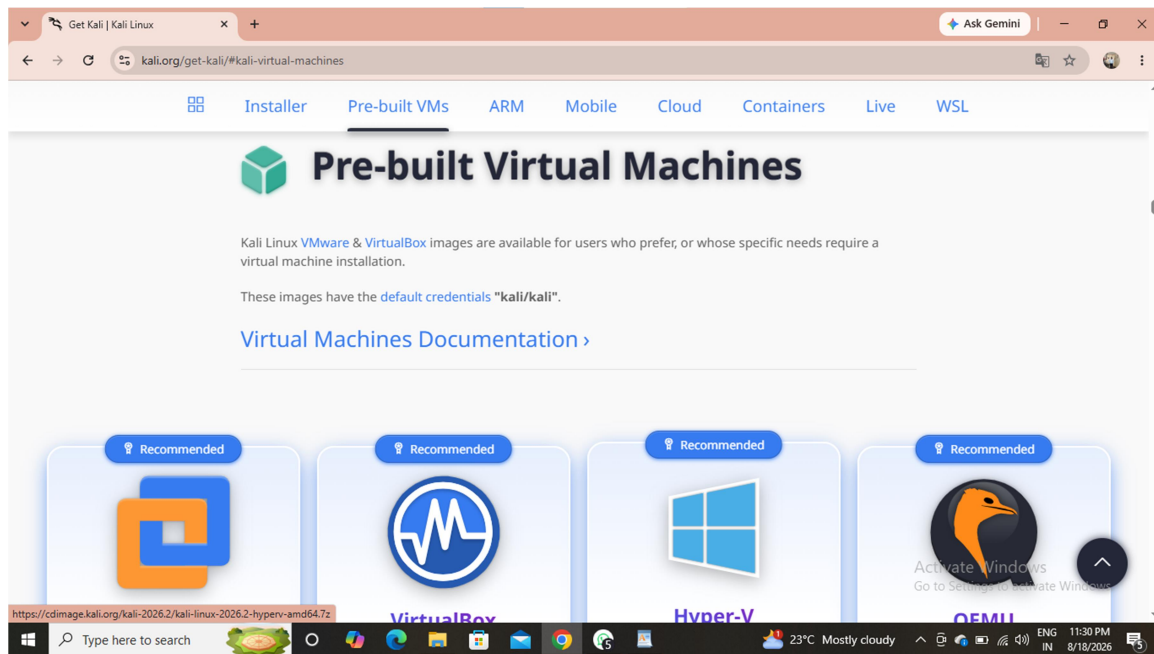
2. Installation of VMM & VM

- 1) Download Oracle VirtualBox for "Windows hosts".
- 2) Download the pre-built Kali Linux VM for VirtualBox.
- 3) Extract the Kali Linux downloaded file.
- 4) Install VirtualBox.
- 5) Open VirtualBox.
- 6) Open the saved/imported VM.
- 7) Start the VM.

STEP 1: Oracle VirtualBox download/installation screen

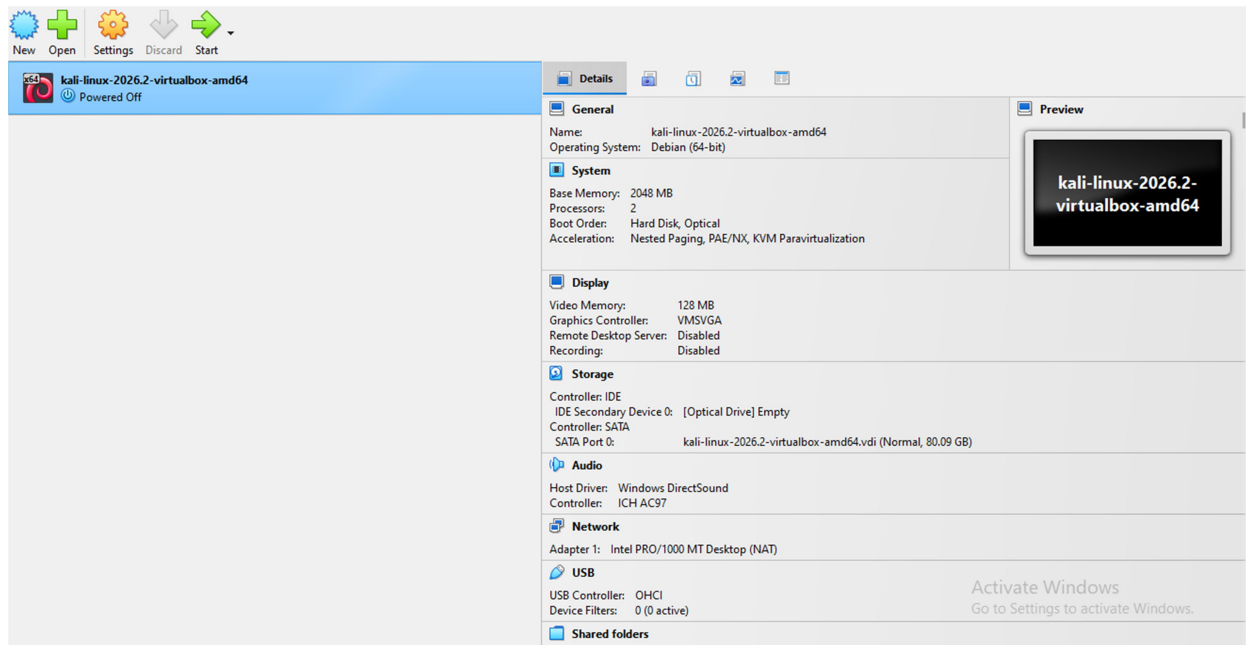


STEP 2: Kali Linux VM downloaded/extracted.



STEP 3: Virtual box main window

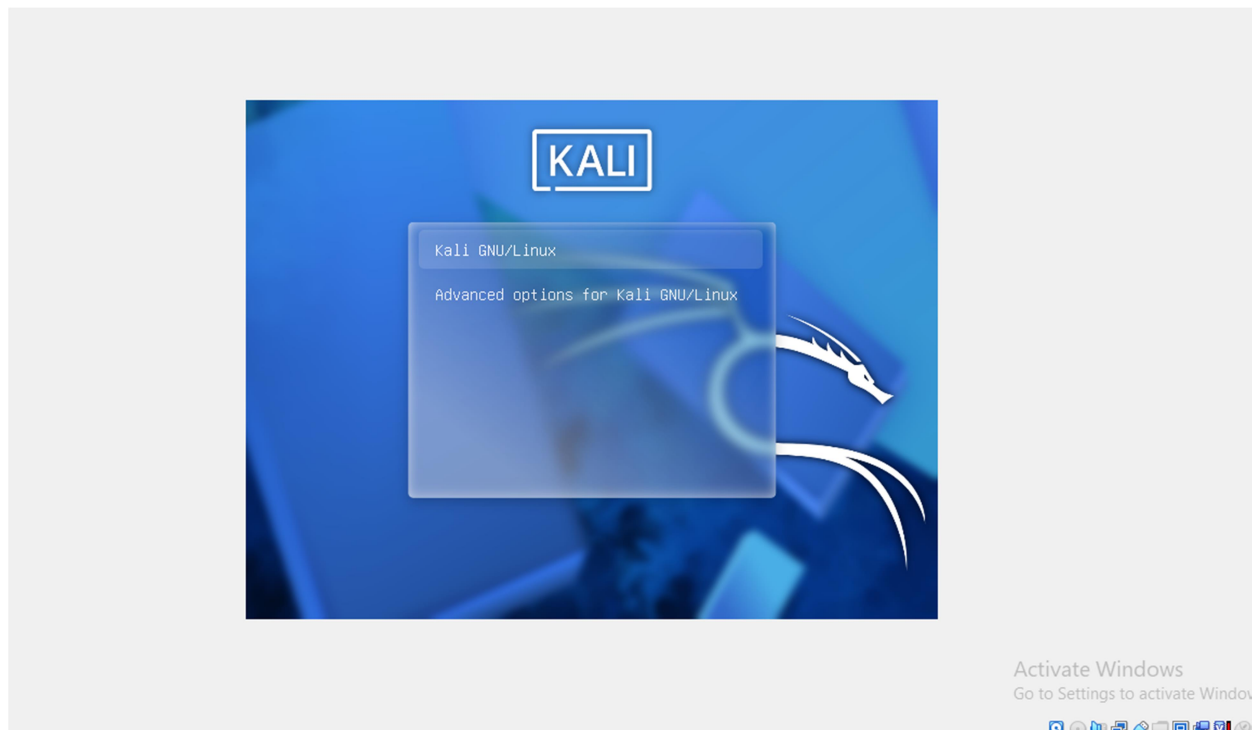




Thus the installation of VMM (Hypervisor Type-2 Oracle virtual box) & VM (Kali Linux) is completed.

STEP 6: Start the VM

st



STEP 7: Simple C program

Open the terminal and create the following simple program, compile and execute to understand the working of the environment.

```
#include <stdio.h>

int main()

{

    printf("Hello World");

    return 0;

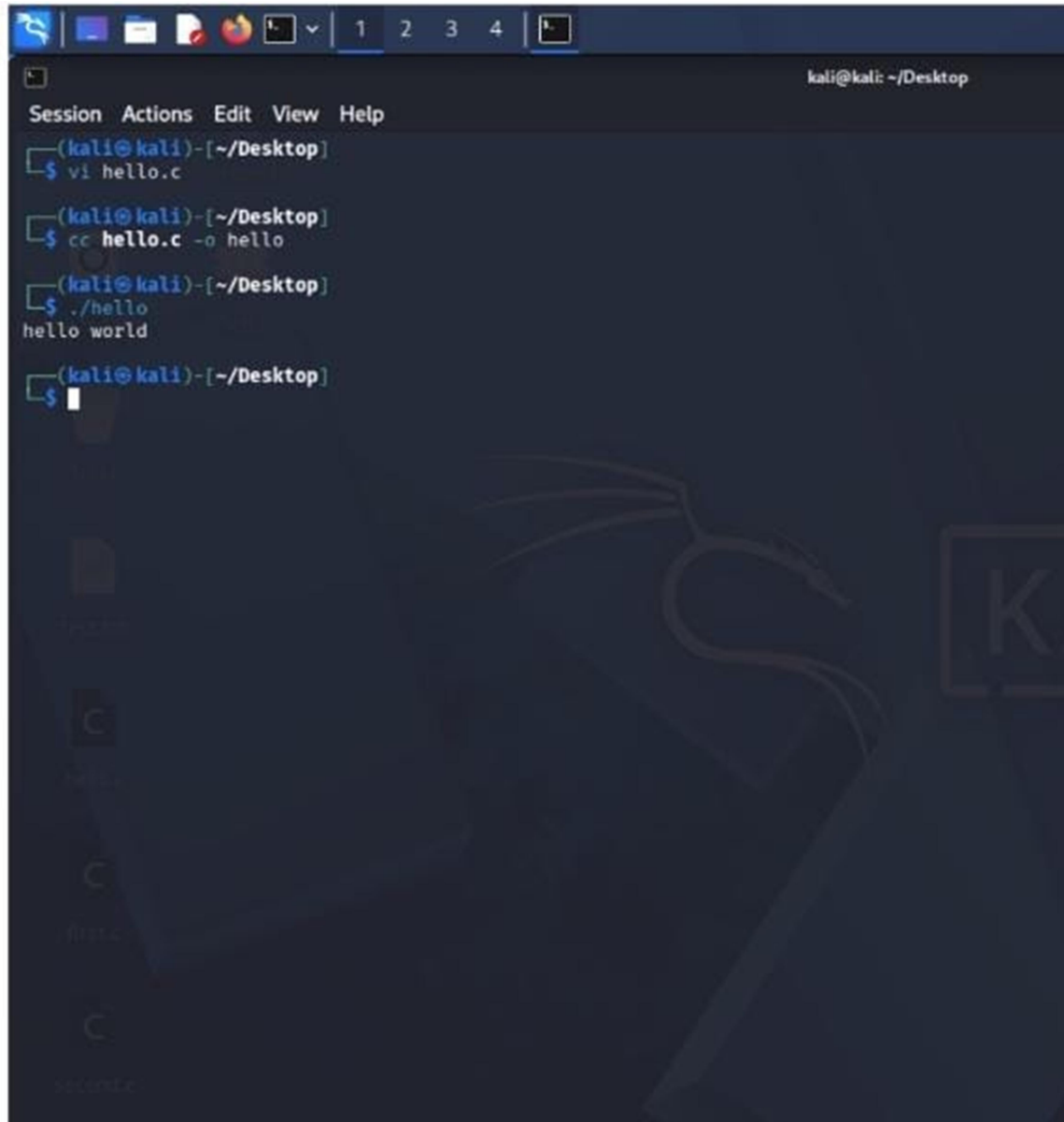
}
```

Compile command: gcc hello.c -o hello

Execution command: ./hello

Expected output: Hello World

Terminal showing the C program, compilation/execution and “Hello World” output.

A screenshot of a Kali Linux terminal window. The window has a dark blue background with a menu bar at the top containing 'Session', 'Actions', 'Edit', 'View', and 'Help'. The terminal shows a series of commands and their outputs. The prompt is '(kali@kali)-[~/Desktop]'. The commands entered are: 'vi hello.c', 'cc hello.c -o hello', and './hello'. The output of the last command is 'hello world'. The terminal also shows a file manager icon on the left side of the window.

```
(kali@kali)-[~/Desktop]
$ vi hello.c
(kali@kali)-[~/Desktop]
$ cc hello.c -o hello
(kali@kali)-[~/Desktop]
$ ./hello
hello world
(kali@kali)-[~/Desktop]
$
```

B) Working with Linux Commands

Execute the following commands in your environment, take output snippets, and provide a brief explanation.

Terminal basics

pwd – Shows the present working directory.

mkdir – Creates a new directory.

rmdir – Removes an empty directory.

dir – Lists files and directories.

ls – Lists files and directories.

cd – Changes the current directory.

cd .. – Moves to the parent directory.

clear – Clears the terminal screen.

history – Shows previously executed commands.

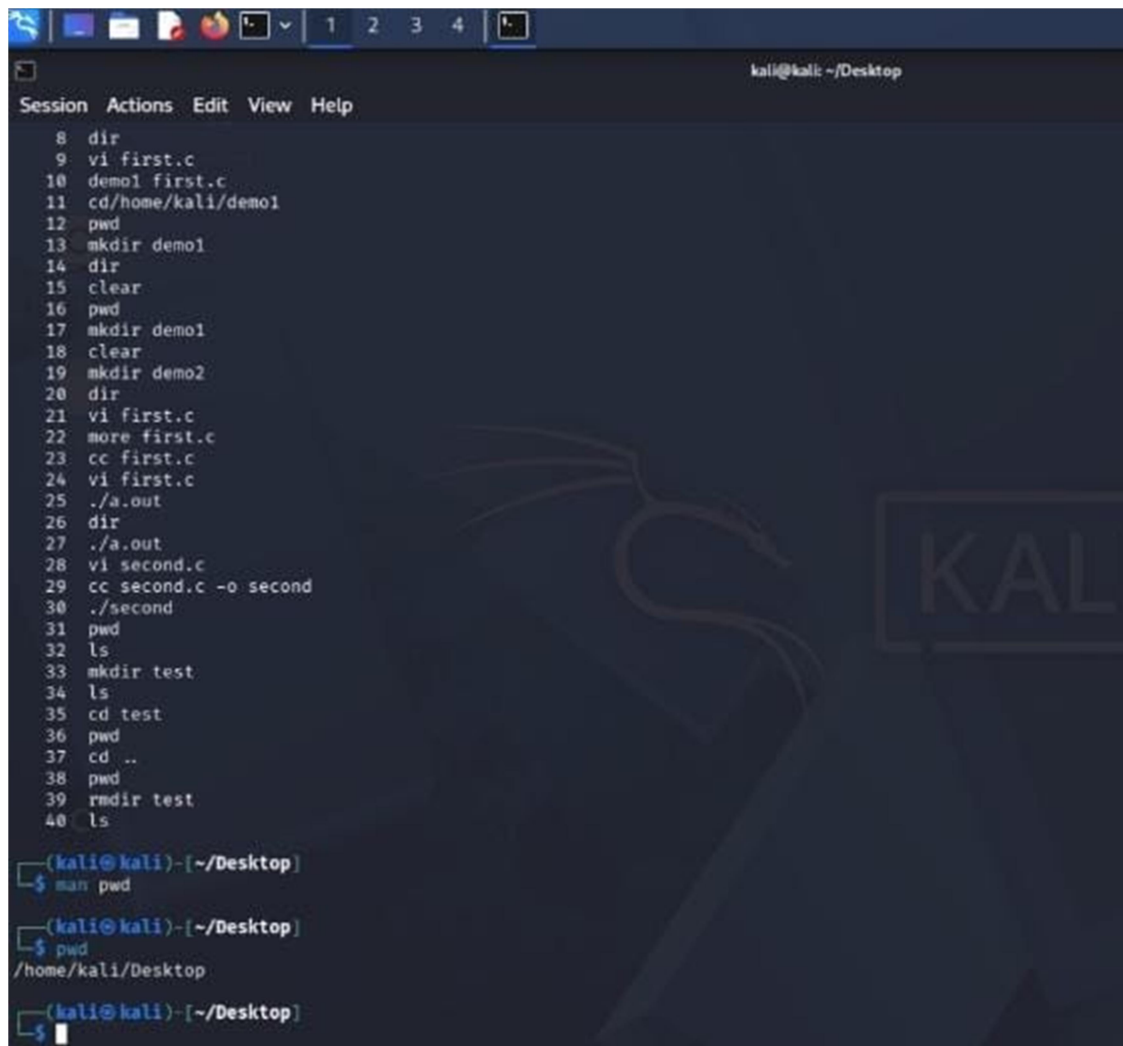
man – Shows the manual/help page of a command.

whoami – Shows the current username.

The terminal output screenshot for the terminal basics commands.

```
kali@kali: ~/Desktop

Session Actions Edit View Help
(kali@kali)~[/Desktop]
$ pwd
/home/kali/Desktop
(kali@kali)~[/Desktop]
$ ls
first.c  second  second.c
(kali@kali)~[/Desktop]
$ mkdir test
(kali@kali)~[/Desktop]
$ ls
first.c  second  second.c  test
(kali@kali)~[/Desktop]
$ cd test
(kali@kali)~[/Desktop/test]
$ pwd
/home/kali/Desktop/test
(kali@kali)~[/Desktop/test]
$ cd ..
(kali@kali)~[/Desktop]
$ pwd
/home/kali/Desktop
(kali@kali)~[/Desktop]
$ rmdir test
(kali@kali)~[/Desktop]
$ ls
first.c  second  second.c
(kali@kali)~[/Desktop]
$ history
1  pwd
2  mkdir
3  mkdir demo1 demo2
4  dir
5  rmdir demo2
```



```
kali@kali: ~/Desktop
Session Actions Edit View Help
8 dir
9 vi first.c
10 demol first.c
11 cd /home/kali/demo1
12 pwd
13 mkdir demo1
14 dir
15 clear
16 pwd
17 mkdir demo1
18 clear
19 mkdir demo2
20 dir
21 vi first.c
22 more first.c
23 cc first.c
24 vi first.c
25 ./a.out
26 dir
27 ./a.out
28 vi second.c
29 cc second.c -o second
30 ./second
31 pwd
32 ls
33 mkdir test
34 ls
35 cd test
36 pwd
37 cd ..
38 pwd
39 rmdir test
40 ls

(kali@kali)-[~/Desktop]
$ man pwd

(kali@kali)-[~/Desktop]
$ pwd
/home/kali/Desktop

(kali@kali)-[~/Desktop]
$
```

Files

vi – Opens the vi text editor.

file – Shows information about a file type.

rm – Removes files/directories.

cp – Copies files/directories.

mv – Moves or renames files/directories.

cat – Displays file contents.

more – Displays contents page by page.

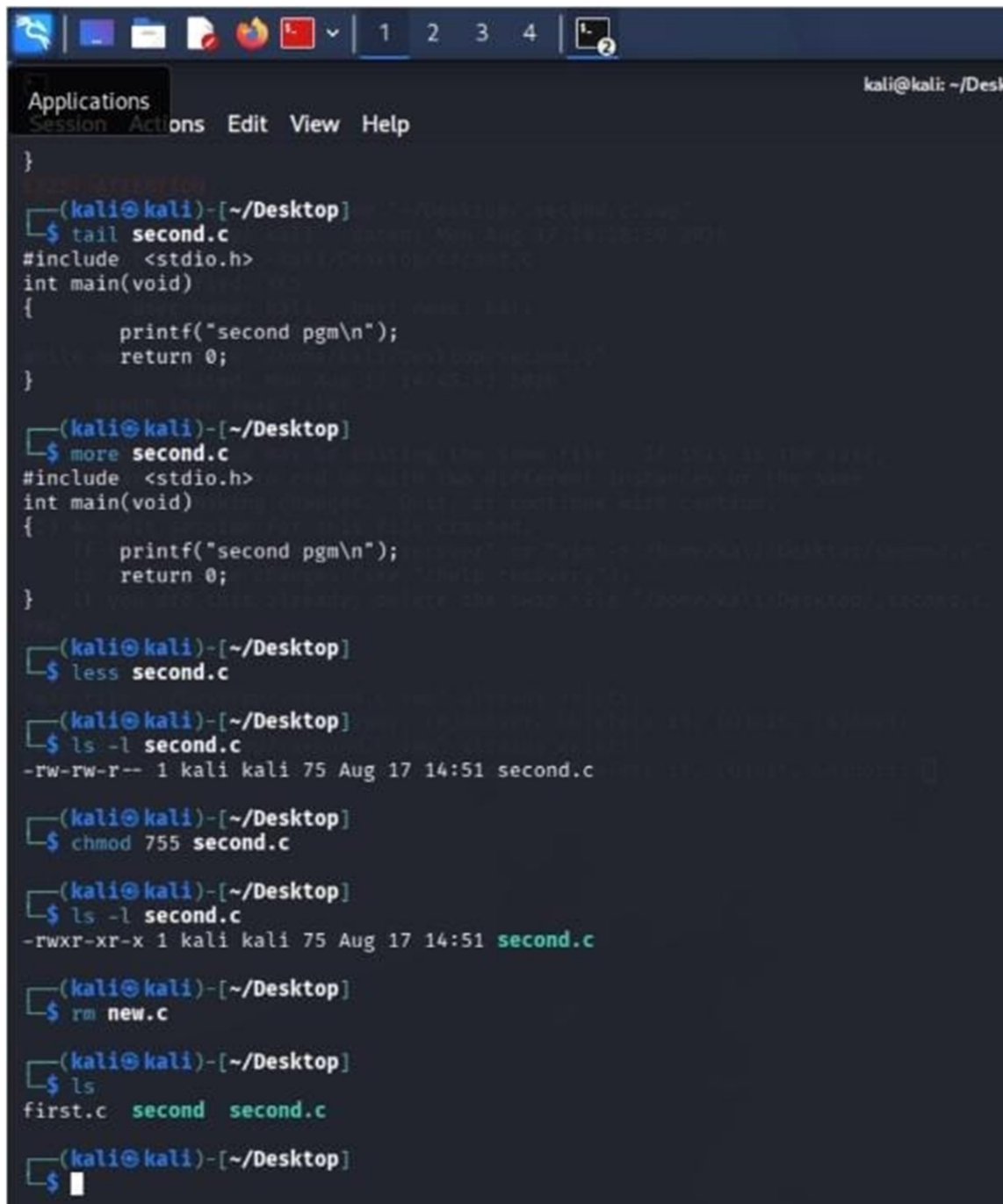
less – Views file contents with navigation.

head – Displays the beginning of a file.

tail – Displays the end of a file.

chmod – Changes file permissions.

The terminal output screenshot for the files command



```
(kali@kali)-[~/Desktop]
└─$ tail second.c
#include <stdio.h>
int main(void)
{
    printf("second pgm\n");
    return 0;
}

(kali@kali)-[~/Desktop]
└─$ more second.c
#include <stdio.h>
int main(void)
{
    printf("second pgm\n");
    return 0;
}

(kali@kali)-[~/Desktop]
└─$ less second.c

(kali@kali)-[~/Desktop]
└─$ ls -l second.c
-rw-rw-r-- 1 kali kali 75 Aug 17 14:51 second.c

(kali@kali)-[~/Desktop]
└─$ chmod 755 second.c

(kali@kali)-[~/Desktop]
└─$ ls -l second.c
-rwxr-xr-x 1 kali kali 75 Aug 17 14:51 second.c

(kali@kali)-[~/Desktop]
└─$ rm new.c

(kali@kali)-[~/Desktop]
└─$ ls
first.c  second  second.c

(kali@kali)-[~/Desktop]
└─$
```

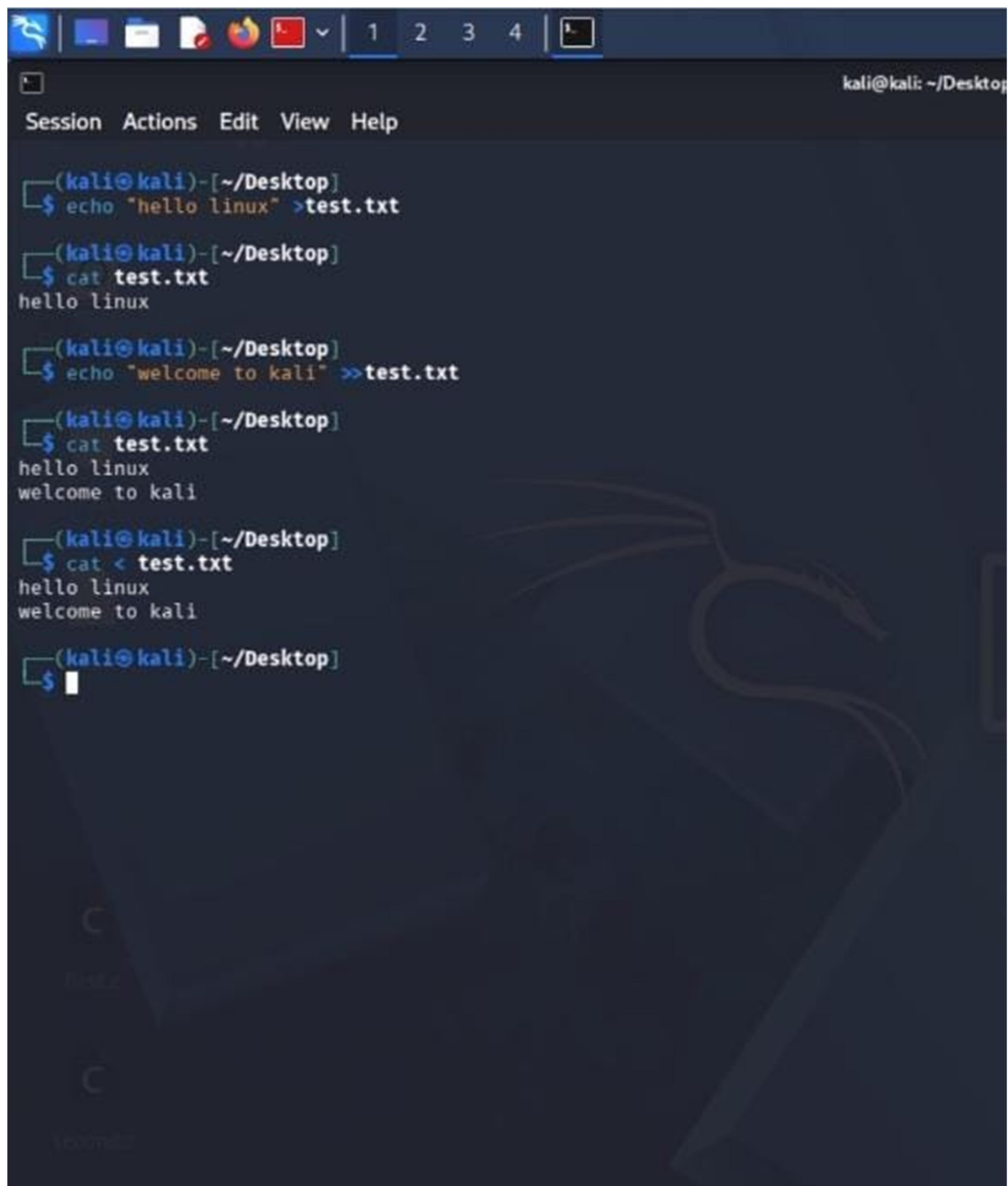
Redirection

> – Redirects output to a file, replacing its contents.

>> – Redirects output and appends it to a file.

< – Takes input from a file.

The terminal output screenshot for the redirection commands.



A terminal window titled "kali@kali: ~/Desktop" with a menu bar (Session, Actions, Edit, View, Help) and a tab bar (1, 2, 3, 4). The terminal shows a series of commands and their outputs:

```
(kali@kali)-[~/Desktop]
$ echo "hello linux" >test.txt

(kali@kali)-[~/Desktop]
$ cat test.txt
hello linux

(kali@kali)-[~/Desktop]
$ echo "welcome to kali" >>test.txt

(kali@kali)-[~/Desktop]
$ cat test.txt
hello linux
welcome to kali

(kali@kali)-[~/Desktop]
$ cat < test.txt
hello linux
welcome to kali

(kali@kali)-[~/Desktop]
$
```

The background of the terminal window features a faint, stylized illustration of a dragon.

Result

The Linux programming environment was setup and the basic linux commands were executed successfully.